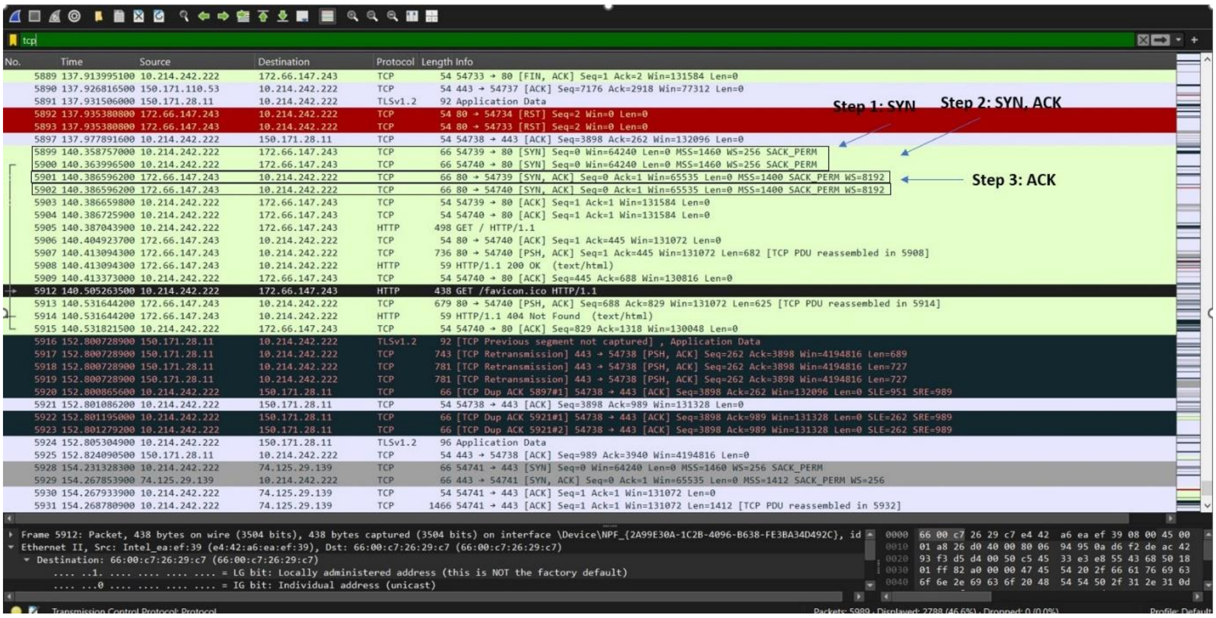
Practical Network Traffic Analysis Using Wireshark

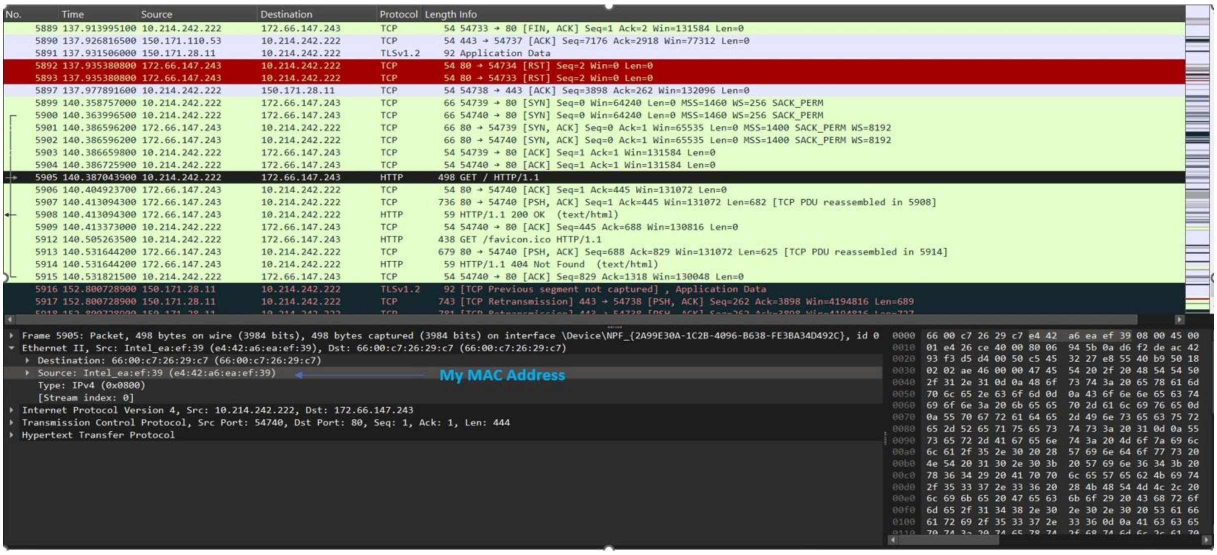
a. TCP 3-way handshake interaction



Explanation:

From my traffic analysis, I clearly noticed that the connection starts with three packets (SYN, SYN-ACK, ACK), and this was very clear around packet number 5899. Basically, the client sends a SYN packet first, then the server replies with SYN-ACK, and finally the client sends an ACK to complete the connection. This matches the explanation of how TCP establishes a reliable connection (Kurose & Ross, 2021).

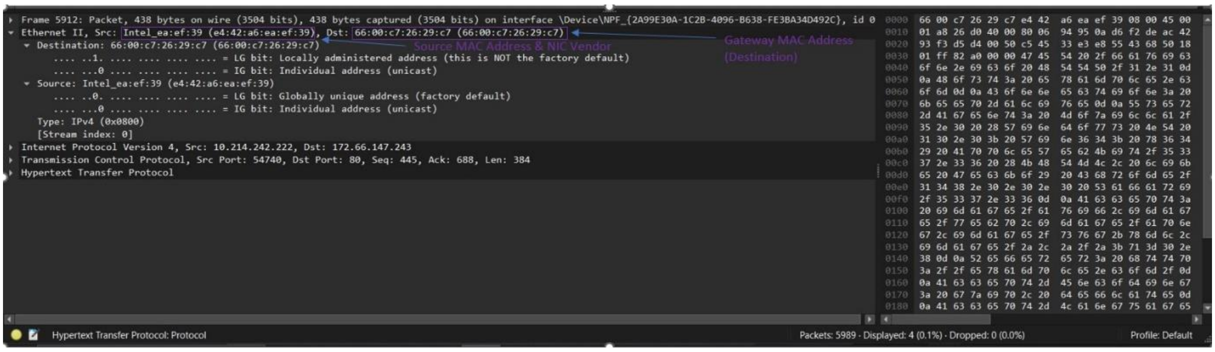
b. MAC address of the PC



Explanation :

While examining the Ethernet frame, I was able to identify my device's MAC address, and I noticed it appears as the source in all outgoing frames. This makes sense because the MAC address is a unique 48-bit identifier assigned to the network interface card (NIC) and operates at the data link layer (Stallings, 2016).

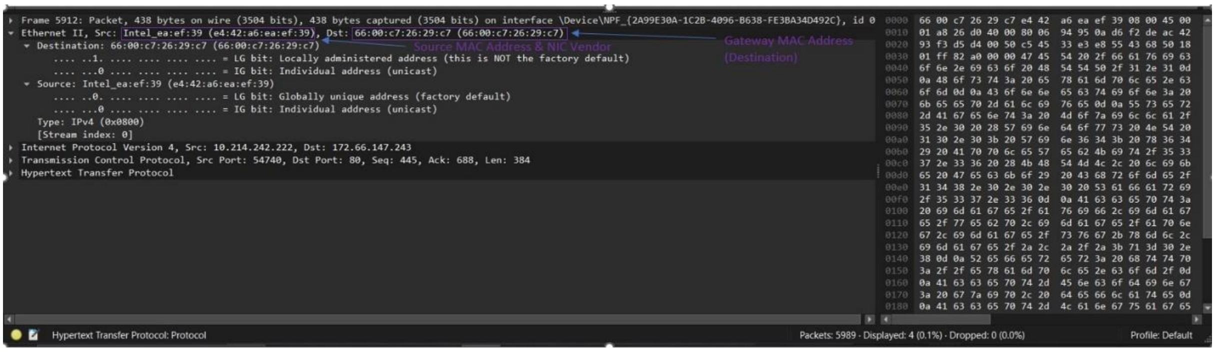
c. Network card vendor/manufacturer



Explanation:

By checking the OUI in Wireshark, I found that the manufacturer of my network card is Intel. From my understanding, the first 24 bits of the MAC address identify the vendor, which ensures global uniqueness of addresses (IEEE, 2023).

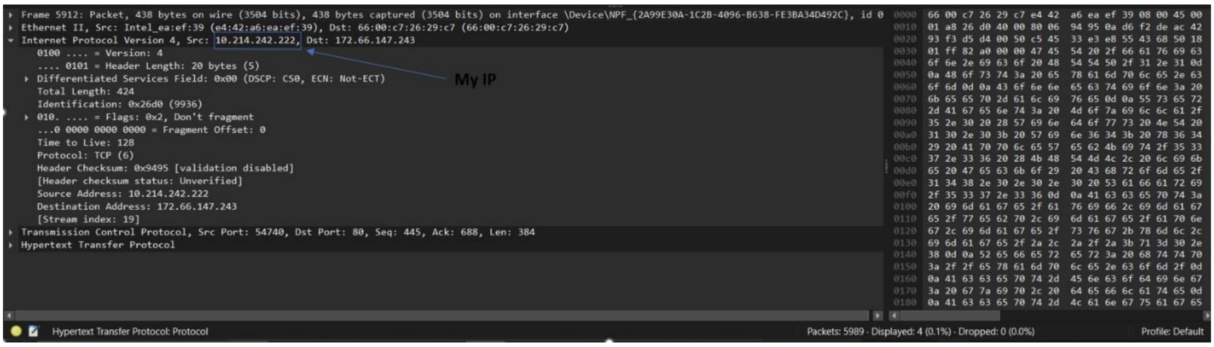
d. MAC address of the gateway device



Explanation:

Since I was accessing a website outside my local network, I noticed that the destination MAC address in outgoing frames belongs to the gateway. This is expected because the gateway forwards traffic from the local network to external networks (Lammle, 2020).

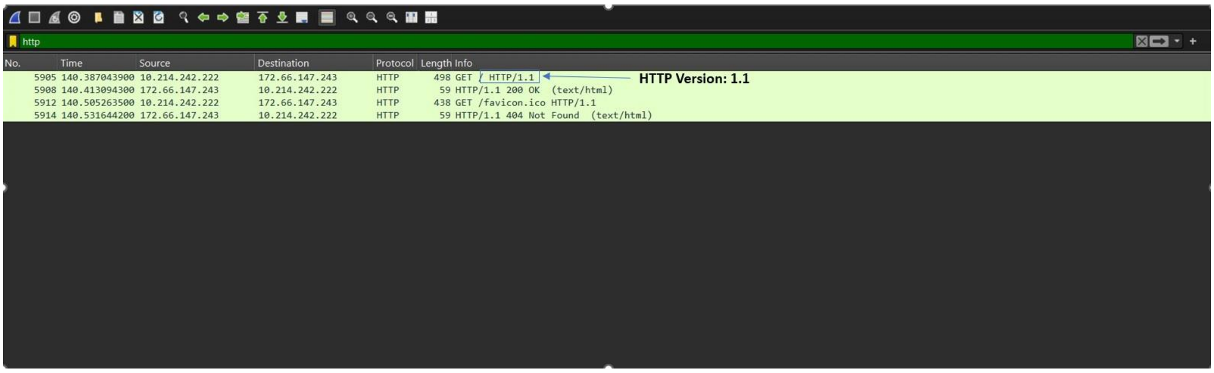
e. The IP address of the PC as shown in Wireshark



Explanation:

I identified my IP address as (10.214.242.222) from the captured packets. From what I understand, the IP address identifies the network interface rather than just the device itself, which aligns with standard networking concepts (Tanenbaum & Wetherall, 2021).

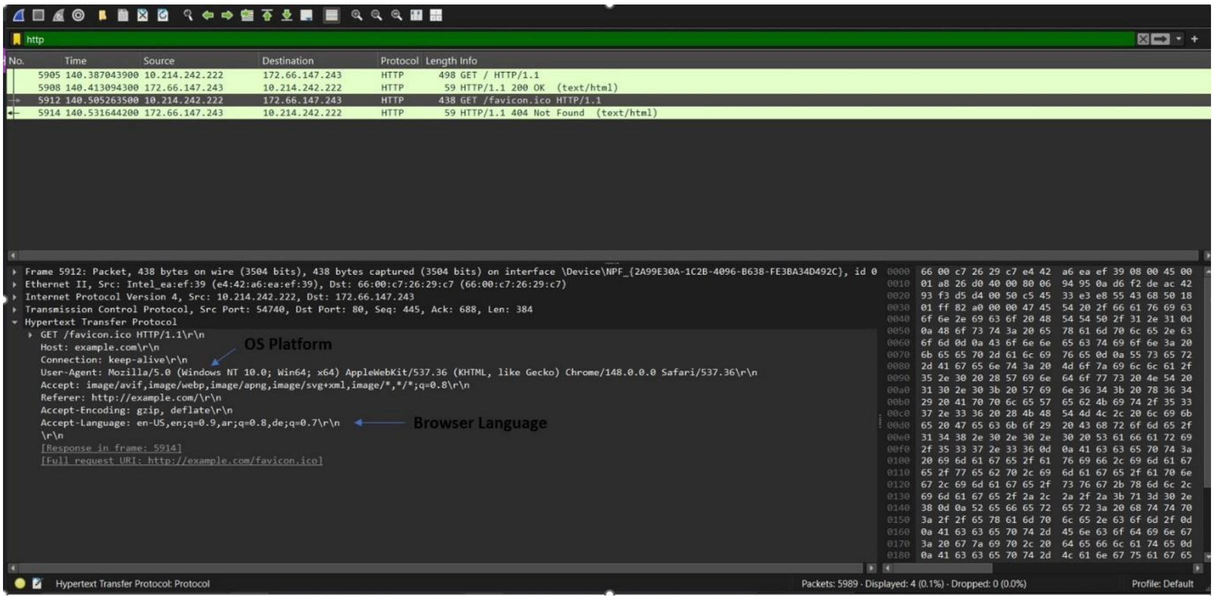
f. Version of HTTP captured



Explanation:

From the captured traffic, I observed that the browser is using HTTP/1.1. This version is still widely used for web communication and supports efficient data transfer between clients and servers (Fielding et al., 1999).

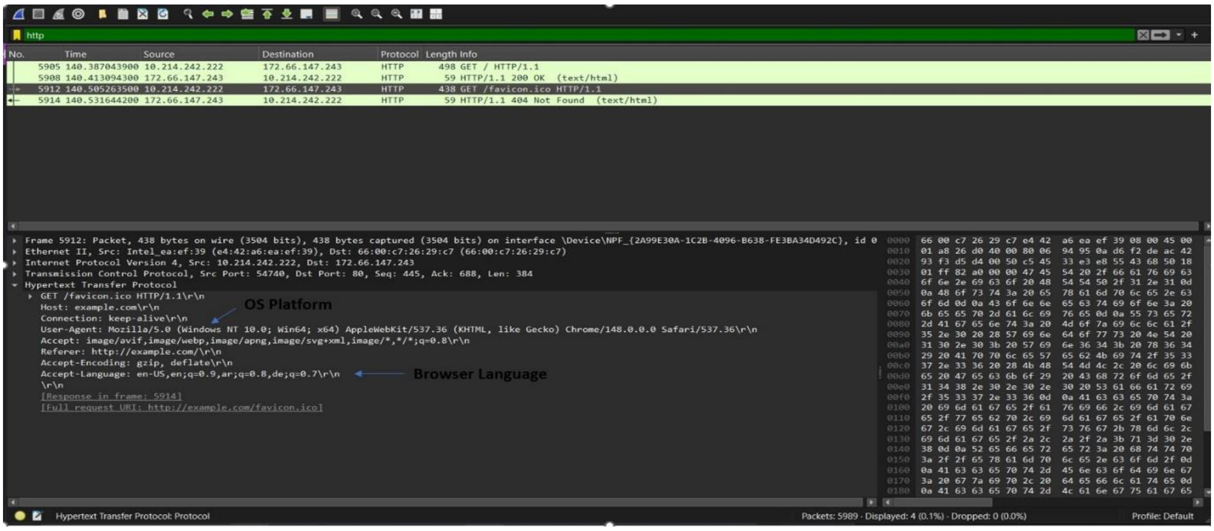
g. PC operating system (OS) platform



Explanation:

By analyzing the User-Agent field, I concluded that the operating system is Windows 10. The User-Agent string provides information about the client's platform and helps servers customize responses (MDN Web Docs, 2024).

h. PC browser language configuration



Explanation:

Finally, I found that the browser language is English (en-US), which was visible in the Accept-Language field. This field allows the browser to indicate preferred languages for content delivery (Reschke & Fielding, 2014).

Reference:

Fielding, R., Gettys, J., Mogul, J., Frystyk, H., Masinter, L., Leach, P., & Berners-Lee, T. (1999). *Hypertext Transfer Protocol -- HTTP/1.1* (RFC 2616). Internet Engineering Task Force. https://www.rfc-editor.org/rfc/rfc2616

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Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach* (8th ed.). Pearson.

Lammle, T. (2020). *CCNA 200-301 Study Guide* (Vol. 1). Sybex.

MDN Web Docs. (2024). *User-Agent header*. https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/User-Agent

Reschke, J., & Fielding, R. (2014). *Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content* (RFC 7231). IETF. https://www.rfc-editor.org/rfc/rfc7231

Stallings, W. (2016). *Data and Computer Communications* (10th ed.). Pearson.

Tanenbaum, A. S., & Wetherall, D. J. (2021). *Computer Networks* (6th ed.). Pearson.